

Which Divergences Control Representations?

Tail Sensitivity and the Limits of Identifiability

Anonymous Author(s)
Anonymous Institution

Working draft — September 25, 2026. Submission target: ICML 2027.

Abstract

Can closeness of output distributions certify that two networks learned similar representations? For the softmax family covering autoregressive language models, contrastive pre-training and supervised classifiers, we show the answer is no in any practically useful sense, and we locate exactly why. Representational error is a variance of normalised log-likelihood ratios, which are largest where probability mass is smallest, whereas every classical divergence weights a label by its own mass. We call the missing property *tail sensitivity* and make the consequence sharp along three independent axes. First, indexing divergences by Rényi order, exact asymptotics give a critical order $\alpha_{\text{sym}}^* > 1$, so the whole segment $\alpha \leq 1$ — total variation, squared Hellinger, Jensen–Shannon, Kullback–Leibler — fails outright. Second, KL’s modulus carries a factor δ_m^{-1} in the conditional-probability floor, and since $p_{(M+1)} \leq 1/(M+1)$ for any model, the bound is empty beyond $\epsilon < 1.7 \times 10^{-3}$; on CIFAR-10 it needs $\epsilon < 10^{-31}$. Both are ceilings, not loose constants. Third, and in contrast, the sup-log-ratio route *does* work once its constant is right: any Frobenius-routed bound obeys bound $\geq \sqrt{2\lambda_{\max}d}$ and so can only certify pairs with $d < 1/(2\lambda_{\max})$, but retrained CIFAR-10 networks satisfy this, and two refinements to the perturbation estimate take the non-vacuity rate on them from 0% to 58% at $M=2$ and 64% at $M=3$, though only 13% at $M=5$ as $\lambda_{\max}$ grows. We complement them with the positive theory that says what would be required: a dichotomy ($\alpha_{\text{sym}}^* = \infty$ exactly for $\sim_L$ -equivalent pairs) and graded control ($d = O(1/\alpha)$), which together confine the controlling orders to an explicit interval. The practical output is therefore a *diagnostic* rather than a certificate: δ_m , $\Delta/\psi_{\min}$, $\lambda_{\max}$ and α_{sym}^* are cheap to compute and bound in advance how much any likelihood comparison can reveal. Experiments on synthetic data and CIFAR-10 confirm the mechanism, and a temperature intervention that holds representational dissimilarity exactly fixed while moving the confidence floor isolates it causally.

1 Introduction

Independently trained networks often appear to learn related representations, and a large literature asks when and why [6, 5]. Any answer depends on what counts as “the same” representation, and identifiability theory supplies a principled choice: for the softmax family of Eq. (1), models with equal likelihood have embeddings equal up to an invertible linear map [1, 2, 3], so a similarity measure should be invariant to exactly that class.

Exact likelihood equality is not what happens in practice, though. Trained models agree *approximately*, and it was recently shown that approximate agreement is not enough: two models can have vanishing Kullback–Leibler divergence and dissimilar representations [5]. It is tempting to read that as a fact about KL, and to look for a better divergence. We argue it is a fact about a *property* that KL happens to lack.

Tail sensitivity. The representational error between two models is a variance of normalised log-likelihood ratios (Sec. 5). Log-ratios are large exactly where probabilities are small, whereas every classical divergence weights label y by

its own probability mass. A divergence therefore controls representations only if it stays sensitive to log-ratios at labels carrying vanishing mass. Call this *tail sensitivity*. The question “which divergences control representations?” becomes “which divergences are tail sensitive?”, and the Rényi family, whose order α tunes precisely this weighting, resolves it.

The answer is an interval. We show the controlling orders form an interval, closed at both ends by theorems. From below, exact asymptotics along norm-growth families give a critical order α_{sym}^* with $\alpha_{\text{sym}}^* > 1$ always (Thm. 5), so the entire segment $\alpha \leq 1$ fails at once: total variation, squared Hellinger, Jensen–Shannon and KL, with the known KL counterexample as the boundary case of a one-parameter statement. From above, a dichotomy (Thm. 21) shows $\alpha_{\text{sym}}^* = \infty$ holds *only* for $\sim_L$ -equivalent pairs, and a graded bound (Thm. 22) turns a large critical order into representational similarity at rate $O(1/\alpha)$. Together they give an explicit cap (Cor. 24), so a finite controlling order provably exists.

Confidence, not quality. The same analysis says when likelihood agreement is informative at all. The modulus for KL scales as δ_m^{-1} , where δ_m is a floor on the conditional probabilities over a diverse label subset (Thm. 15). Confident classifiers and language models place negligible mass on most labels, so δ_m is tiny and matched likelihood tells one almost nothing about matched representations — a prediction we confirm on CIFAR-10, and reverse by label smoothing.

Contributions.

- *Control* (Def. 1), making divergence–representation claims falsifiable and uniform over a model class.
- Exact rate asymptotics and a closed-form critical order (Thm. 5), verified to solver precision; hence every order $\alpha \leq 1$ fails (Cor. 6).
- A perturbation bound $d_{\mathbf{f},\mathbf{g}} \leq 2\sqrt{\lambda_{\max}\rho_2/\psi_{\min}}$ (Thm. 11), improving the known modulus by $\sqrt{M}$ via Hoffman–Wielandt (Prop. 10).
- A dichotomy (Thm. 21), graded control (Thm. 22), and an explicit finite cap on the controlling order (Cor. 24).
- Experiments on constructed models, synthetic angular classification and CIFAR-10, including a label-smoothing intervention that restores KL’s predictive power as the theory predicts.

2 Setup

Models are pairs $(\mathbf{f}, \mathbf{g}) \in \Theta$ with embedding $\mathbf{f} : \mathcal{X} \rightarrow \mathbb{R}^M$ and unembedding $\mathbf{g} : \mathcal{Y} \rightarrow \mathbb{R}^M$, inducing

$$p_{\mathbf{f},\mathbf{g}}(y | x) = \frac{\exp(\mathbf{f}(x)^\top \mathbf{g}(y))}{\sum_{y' \in \mathcal{Y}} \exp(\mathbf{f}(x)^\top \mathbf{g}(y'))}. \quad (1)$$

Write $\mathbf{f}_0(x) = \mathbf{f}(x) - \mathbf{f}(x_0)$, $\mathbf{g}_0(y) = \mathbf{g}(y) - \mathbf{g}(y_0)$; let $\mathbf{L}$ have columns $\mathbf{g}_0(y)$ and $\mathbf{N}$ columns $\mathbf{f}_0(x)$. Under a diversity condition, equal likelihoods imply $\mathbf{f} = \mathbf{A}\mathbf{f}'$ and $\mathbf{g}_0 = \mathbf{A}^{-\top} \mathbf{g}'_0$ for invertible $\mathbf{A}$ [1, 2, 3].

3 When Does a Divergence Control Representations?

Definition 1 (Control). A divergence D controls $d_{\mathbf{f},\mathbf{g}}$ on a model class Θ if there is a nondecreasing $h : \mathbb{R}_+ \rightarrow \mathbb{R}_+$ with $h(0^+) = 0$ such that for all $(\mathbf{f}, \mathbf{g}), (\mathbf{f}', \mathbf{g}') \in \Theta$,

$$D(p_{\mathbf{f},\mathbf{g}}, p_{\mathbf{f}',\mathbf{g}'}) \leq \epsilon \implies d_{\mathbf{f},\mathbf{g}}((\mathbf{f}, \mathbf{g}), (\mathbf{f}', \mathbf{g}')) \leq h(\epsilon). \quad (2)$$

We call h a *modulus of control*.

Remark 2. The quantifiers matter: control is uniform over Θ . A divergence may fail on Θ yet control on a subclass, which is exactly the situation for KL (Sec. 5).

Lemma 3 (Transfer). *If D_2 controls with modulus h_2 and there is increasing ϕ with $\phi(0^+) = 0$ and $D_2 \leq \phi(D_1)$ on Θ , then D_1 controls with $h_1 = h_2 \circ \phi$.*

Because D_α is nondecreasing in α , Lem. 3 makes the set of controlling orders upward closed, so $\alpha_\Theta^* := \inf\{\alpha : D_\alpha \text{ controls } d_{\mathbf{f},\mathbf{g}}\}$ is well posed. The rest of the paper brackets it.

4 Tail Insensitivity and Impossibility

4.1 The co-concentration mechanism

Fix x , write $s_y = \cos(\mathbf{f}(x), \mathbf{g}(y))$ and $\hat{y} = \arg \max_y s_y$, and define the *margin gaps*

$$a_y := s_{\hat{y}} - s_y > 0, \quad b_y := s'_{\hat{y}} - s'_y > 0. \quad (3)$$

Under the norm-growth family, logits are $c\rho s_y$, so $p(y | x) \propto e^{-c\rho a_y}$ and $p'(y | x) \propto e^{-c\rho b_y}$.

Lemma 4 (Co-concentration). *Both models converge to the same point mass: $p(\hat{y} | x) \rightarrow 1$ and $p'(\hat{y} | x) \rightarrow 1$, hence $\|p - p'\|_{\text{TV}} \rightarrow 0$, while $\sup_y |\log p - \log p'| = \Theta(\rho)$.*

4.2 The rate lemma

Define the *tilted exponent*

$$E_\alpha(y) := \alpha a_y + (1 - \alpha) b_y = b_y + \alpha(a_y - b_y). \quad (4)$$

Theorem 5 (Rate lemma and critical order). *Let $\mathcal{P} := \{y \neq \hat{y} : a_y \neq b_y\} \neq \emptyset$ and put*

$$\alpha^* := \min_{y : b_y > a_y} \frac{b_y}{b_y - a_y}. \quad (5)$$

Then $\alpha^ > 1$, and as $\rho \rightarrow \infty$:*

- for $\alpha < \alpha^*$, $D_\alpha = \Theta(e^{-c\rho\kappa(\alpha)})$, where $\kappa(\alpha)$ is the smallest value in $\{E_\alpha(y)\} \cup \{a_y\} \cup \{b_y\}$ carrying a non-vanishing total coefficient — labels with $a_y = b_y$ cancel exactly and are excluded;*
- at $\alpha = 1$ all first-order coefficients cancel and $d_{\text{KL}} = \Theta(\rho e^{-c\rho \min_{\mathcal{P}} a_y})$;*
- for $\alpha > \alpha^*$, D_α grows linearly in ρ with slope $\max_y (-E_\alpha(y)) / (\alpha - 1)$, increasing to $\max_y (b_y - a_y)$ as $\alpha \rightarrow \infty$.*

Proof sketch. Expand $\log \sum_y p^\alpha p'^{1-\alpha}$ to first order and group the three exponent families by value; at value v the total coefficient is $\#\{E_\alpha = v\} - \alpha\#\{a = v\} + (\alpha - 1)\#\{b = v\}$. Labels with $a_y = b_y$ give $E_\alpha(y) = a_y = b_y$ and contribute $1 - \alpha + (\alpha - 1) = 0$. At $\alpha = 1$ every coefficient vanishes, forcing the second-order term. Finally $0 < b_y - a_y < b_y$ gives $\alpha^* > 1$. $\square$

Corollary 6 (The classical divergences sit below threshold). *Since $\alpha^* > 1$, the family defeats every divergence of order $\alpha \leq 1$: total variation, squared Hellinger, Jensen–Shannon and Kullback–Leibler all fail to control $d_{\mathbf{f},\mathbf{g}}$. The known KL result is the boundary case $\alpha = 1$ of a one-parameter statement.*

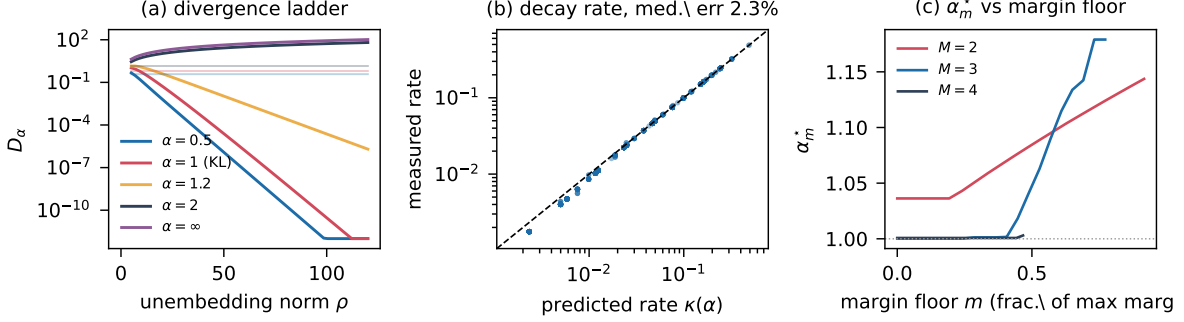


Figure 1: (a) Divergence ladder along the norm-growth family at a single input: orders $\alpha \leq 1$ (including KL) decay, orders above α^* grow linearly. Faint horizontal lines are $d_{f,g}$, m_{CCA} and d_{LLV}^λ , constant to four digits throughout. (b) Predicted decay rate $\kappa(\alpha)$ against the measured exponential rate, 739 (geometry, order) pairs spanning 2.3 decades; median relative error 2.3%. This is the check that exercises the cancellation rule of Thm. 5(i): omitting the $a_y = b_y$ exclusion inflates the error to 40–70%. (c) Margin-conditioned threshold $\alpha_m^* := \min\{\alpha^*(x) : \text{margin}(x) \geq m\}$, where $\text{margin}(x) = \min_{y \neq y} a_y$, for $M = 2, 3, 4$. Conditioning on a margin floor removes the boundary inputs that pin the unconditioned threshold at 1, and makes explicit that α^* is set by the worst-margin input rather than by the representational geometry.

4.3 A pointwise trade-off

Proposition 7 (Trade-off, fixed input). *Fix x . Then $\alpha^*(x)$ is large only when $a_y \approx b_y$ at the critical label, which simultaneously drives the pointwise signal $\max_y(b_y - a_y)$ toward zero.*

Proof. $\square$

Remark 8 (The trade-off does *not* survive averaging). Prop. 7 is genuinely pointwise and must not be read as a statement about model pairs. Over an input distribution the correlation between $\alpha^* = \min_x \alpha^*(x)$ and $d_{f,g}$ falls to -0.16 , and the input-concentration knob moves the two the *same* way: tightening the clusters raises α^* from 1.013 to 1.234 while $d_{f,g}$ rises slightly, $0.665 \rightarrow 0.701$. The mechanism is Fig. 1(c): α^* is a minimum over inputs, so it is pinned by whichever input lies closest to a decision boundary, independently of how dissimilar the representations are. We report this as a negative control (trackA/tradeoff.py).

Theorem 9 (Uniform separation). *There is $c > 0$, independent of ρ and explicit in the cluster geometry, with $\max d_{SVD} \geq c$ for every member of the family.*

Proof. $\square$

5 Which Divergences Do Control

Everything positive follows from one perturbation bound. Let $z_1 := \mathbf{L}^\top \mathbf{f}(x)$ and $z_2 := \mathbf{L}'^\top \mathbf{f}'(x)$, whose l -th entries are $\log p(y_l|x) - \log p(y_0|x)$ and its primed analogue. Writing $r_y(x) := \log p(y|x) - \log p'(y|x)$ and $\Delta_l := z_{1,l} - z_{2,l} = r_{y_l} - r_{y_0}$, set

$$\rho_2 := \max_l \text{std}_x[\Delta_l], \quad \psi_{\min} := \min_l \min\{\text{std}(z_{1,l}), \text{std}(z_{2,l})\} \quad (6)$$

Note $\text{std}(z_{1,l}) = \psi_x(y_l; p)$, so $\psi_{\min}$ is exactly the non-degeneracy already present in the LLV construction.

Proposition 10 (Tighter spectral step). *Hoffman–Wielandt gives $\sum_i (\sigma_i - \tilde{\sigma}_i)^2 \leq \|E\|_F^2$, so by Cauchy–Schwarz $\frac{1}{M} \sum_i |\sigma_i - \tilde{\sigma}_i| \leq \|E\|_F / \sqrt{M}$. Combined with the trace bound $\|E\|_F^2 \leq M \sum_l \text{Var}[z'_l - w'_l]$ this improves Lem. E.7 from $d_{SVD} \leq (M \sum_l \text{Var})^{1/2}$ to*

$$d_{SVD} \leq \left(\sum_l \text{Var}[z'_l - w'_l] \right)^{1/2}, \quad (7)$$

a factor $\sqrt{M}$.

Theorem 11 (Perturbation bound). *Under the non-degeneracy conditions of Sec. 2,*

$$d_{f,g} \leq \sqrt{\lambda_{\max}} Q, \quad Q^2 := \frac{1}{M} \sum_l \left(\frac{\tau_l + \text{sd}_l}{\psi_l} \right)^2, \quad (8)$$

where $\lambda_{\max} := \lambda_{\max}(\text{Corr}(z_1)) \leq M$, $\text{sd}_l = \text{std}(\Delta_l)$, $\tau_l = |s_l - s'_l|$ and $\psi_l = \min(s_l, s'_l)$. In particular $D_\infty \leq \epsilon$ gives $Q \leq 4\epsilon/\psi_{\min}$ and hence $d_{f,g} \leq 4\sqrt{\lambda_{\max}}\epsilon/\psi_{\min}$.

Proof. Write $s_l = \text{std}(z_{1,l})$, $s'_l = \text{std}(z_{2,l})$, $\text{sd}_l = \text{std}(\Delta_l)$ and $\tau_l = |s_l - s'_l|$. Since $z_{2,l} = z_{1,l} - \Delta_l$,

$$z'_l - w'_l = (z_{1,l} - \mu_l) \left(\frac{1}{s_l} - \frac{1}{s'_l} \right) + \frac{\Delta_l - \mathbb{E}\Delta_l}{s'_l},$$

whose two terms have standard deviations τ_l/s'_l and sd_l/s'_l , giving $\text{std}[z'_l - w'_l] \leq (\tau_l + \text{sd}_l)/\psi_l$ with $\psi_l = \min(s_l, s'_l)$. For the spectral step, $\|E\|_F^2 = \sum_j \sum_k \text{Cov}[z'_k, \Delta'_j]^2$, whose inner sum is the squared norm of the projection of Δ'_j onto $\text{span}\{z'_k\}$ and so at most $\lambda_{\max} \text{Var}[\Delta'_j]$, rather than the $M \text{Var}[\Delta'_j]$ given by the entrywise covariance inequality. With (7) this yields (8). The same argument on $\mathbf{N}^\top \mathbf{g}(y)$ bounds the unembedding term. $\square$

Remark 12 (Two refinements, and where the rest of the slack sits). Eq. (8) differs from a naive estimate in two ways, each free. It aggregates the coordinates in root mean square rather than taking the worst one ($1.35\text{--}1.57\times$ measured), and it keeps τ_l separate instead of bounding it by sd_l , which is wasteful because the difference of standard deviations is much smaller than the standard deviation of the difference: median τ_l/sd_l is $0.25\text{--}0.53$. Jointly they tighten the bound by $1.65\text{--}2.37\times$ ($1.79\times$ at $M = 2, 3$).

Decomposing further shows the residual slack is *not* spectral. In standardised coordinates $\text{Var}[z'_l - w'_l] = 2(1 - \rho_l)$ exactly, so $d_{\text{SVD}} \leq \sqrt{2\lambda_{\max}(1 - \bar{\rho})}$, and this form sits within 11–17% of the Prop. 13 floor. All remaining looseness is therefore in passing from the log-ratio perturbation to the coordinate correlations ρ_l , which is the precise open problem.

Proposition 13 (Non-vacuity is structurally capped). *In standardised coordinates $\text{Var}[z'_l - w'_l] = 2(1 - \rho_l)$ with $\rho_l = (\Sigma_{z'w'})_{ll}$, so $\sum_l \text{Var}[z'_l - w'_l] = 2M - 2\text{tr}(\Sigma_{z'w'})$. Since $\text{tr} \leq \sum_i \sigma_i = M(1 - d_{\text{SVD}})$,*

$$\sum_l \text{Var}[z'_l - w'_l] \geq 2M d_{\text{SVD}}. \quad (9)$$

The bound of Thm. 11 therefore satisfies bound $\geq \sqrt{2\lambda_{\max}d_{\text{SVD}}}$, and

$$\text{bound} < 1 \implies d_{\text{SVD}} < \frac{1}{2\lambda_{\max}} \leq \frac{1}{2}. \quad (10)$$

Remark 14 (What the cap means). Thm. 11 can only certify pairs that are *already* similar, and the 5.3% non-vacuity rate of Sec. 9 is not a loose constant awaiting a sharper argument: it is the fraction of pairs falling below a hard ceiling. On constructed models the ceiling is $0.16\text{--}0.28$ for $M = 2, \dots, 6$ and *no* pair cleared it. Eq. (9) is an identity-level fact, so the cap binds any argument routed through $\sum_l \text{Var}$, that is through a Frobenius-norm perturbation estimate. Escaping it requires bounding the singular values of $\Sigma_{z'w'}$ directly rather than through $\|E\|_F$. We regard that as the most valuable open technical problem raised here.

Theorem 15 (KL controls on Θ_δ). *Let $\mathcal{Y}_m \subseteq \mathcal{Y}$ be any label subset with $|\mathcal{Y}_m| \geq M + 1$ on which diversity holds, take $\mathcal{Y}_{\text{LLV}} \subseteq \mathcal{Y}_m$, and set $\delta_m := \min_x \min_{y \in \mathcal{Y}_m} \min\{p(y|x), p'(y|x)\}$. If $d_{\text{KL}} \leq \epsilon$ then*

$$d_{\text{f,g}} \leq \frac{4\sqrt{2\lambda_{\max}} \sqrt{\epsilon}}{\delta_m \psi_{\min}}. \quad (11)$$

Remark 16 (Why a subset, and how much it buys). Only labels in $\mathcal{Y}_{\text{LLV}}$ enter z_1, z_2 , so the floor need only be taken over $\mathcal{Y}_m$; no tail-mass term appears, and the sole constraint is $|\mathcal{Y}_m| \geq M + 1$. The gain is real but far too small. On a ResNet18 CIFAR-10 model with $M = 2$ (test accuracy 0.747, NLL 0.753), the operative floor is

$\delta_3 = 2.81 \times 10^{-15}$, so (11) falls below 1 only for $\epsilon < 1.2 \times 10^{-31}$ — some thirty-one orders of magnitude below the KL actually observed between retrained models. The subset reformulation does *not* rescue Thm. 15 for trained classifiers.

The reason, however, is not generic over-confidence. That model is mild: mean maximum probability 0.717, mean predictive entropy 0.685 nats against $\log 10 = 2.303$ for the uniform, and mean perplexity 2.416, i.e. roughly two and a half live classes per input. Its *median* third-largest probability is 2.1×10^{-2} , thirteen orders of magnitude above δ_3 . The floor is destroyed by a handful of outlier inputs among 10,000, not by the typical one.

Proof. For $y \in \mathcal{Y}_m$, $|r_y| = |\log(p_y/p'_y)| \leq |p_y - p'_y|/\delta_m \leq \|p - p'\|_1/\delta_m$. Pinsker gives $\|p - p'\|_1^2 \leq 2D_{\text{KL}}(p\|p')$, so $\mathbb{E}_x[r_y^2] \leq 2\epsilon/\delta_m^2$. Then $\rho_2 \leq 2\max_y(\mathbb{E}_x r_y^2)^{1/2} \leq 2\sqrt{2\epsilon}/\delta_m$, and Thm. 11 concludes. $\square$

Corollary 17 (Confidence, not quality). *The modulus degrades as δ_m^{-1} , and Rem. 16 shows the subset fix recovers only about one order of magnitude. For a confident classifier small KL divergence therefore carries almost no representational information, whatever label subset is chosen. Raising δ_m requires changing the model, not the analysis — which is what the label-smoothing intervention of Sec. 9 tests.*

Remark 18 (The floor is not the bottleneck). It is tempting to blame the minimum over inputs and replace it by a quantile. Measured on the CIFAR-10 model of Rem. 16, discarding the worst 1% of inputs raises the floor to 8.5×10^{-12} and the worst 10% to 2.1×10^{-6} — nine orders of magnitude, and gradual rather than a cliff. It is not enough. Since (11) is below 1 only for $\epsilon < \delta_m^2/64$, the 10%-quantile floor still demands $\epsilon < 6.8 \times 10^{-14}$.

The obstruction is structural. The $(M+1)$ -th largest entry of a probability vector obeys $p_{(M+1)} \leq 1/(M+1)$, so for $M = 2$ we have $\delta_3 \leq 1/3$ for *any* model, and (11) can never be non-vacuous beyond $\epsilon < 1/576 \approx 1.7 \times 10^{-3}$. No choice of $\mathcal{Y}_m$ or quantile repairs this. The reason is the subject of this paper: d_{KL} is a p -weighted average of the log-ratios whereas ρ_2 is their unweighted spread, and converting one into the other must pay the ratio between the weighting and the uniform measure, which is precisely δ_m^{-1} . Tail insensitivity, written as an inequality.

We stress that the *phenomenon* remains measurable where the *bound* is empty: raising δ_m by label smoothing lifts the rank correlation between d_{KL} and $d_{\text{f,g}}$ from 0.550 to 0.821 (Tab. 4), well inside the regime where Thm. 15 asserts nothing.

Lemma 19 (Rényi dominates the sup log-ratio). *For $\alpha > 1$ and $\delta := \min_{x,y} p(y|x)$,*

$$\max_y r_y \leq D_\alpha(p||p') + \frac{\log(1/\delta)}{\alpha - 1}. \quad (12)$$

Proof. Bound the sum by its largest term and use $\alpha \log p + (1 - \alpha) \log p' = \log p + (\alpha - 1)r_y$. $\square$

Remark 20 (The two axes are one). Control requires $\alpha - 1 \gtrsim \log(1/\delta)$: the Rényi order and the confidence floor trade off directly. Thm. 11 is the $\alpha \rightarrow \infty$ endpoint, Thm. 15 the $\alpha = 1$ one.

6 A Dichotomy, and Graded Control

Rényi divergence is asymmetric, so the meaningful critical order is $\alpha_{\text{sym}}^* := \min\{\alpha^*(p||p'), \alpha^*(p'||p)\}$: a family is a counterexample only if *both* directions stay small.

Theorem 21 (Dichotomy). *For $(f, g), (f', g') \in \Theta$ satisfying diversity,*

$$\alpha_{\text{sym}}^* = \infty \iff p_{f,g} = p_{f',g'} \iff (f, g) \sim_L (f', g'). \quad (13)$$

Every non-equivalent pair has a finite critical order.

Proof. $\alpha^*(p||p') = \infty$ means no y has $b_y > a_y$ and $\alpha^*(p'||p) = \infty$ means no y has $a_y > b_y$; together $a_y(x) = b_y(x)$ for all x, y . With $R_y(x) := f(x)^\top g(y) - f'(x)^\top g'(y)$ and $a_y - b_y = R_{\hat{y}} - R_y$, this says $R_y(x) = c(x)$ is independent of y ; then $c(x)$ cancels in the softmax normaliser, so the conditionals coincide, and linear identifiability gives $\sim_L$ -equivalence. The converse is immediate. $\square$

Theorem 22 (Graded control). *Let $\Delta := \sup_{x,y} a_y(x)$. If $\alpha_{\text{sym}}^* \geq A$ then $\sup_{x,l} |\Delta_l| \leq 2\Delta/A$, hence $\rho_2 \leq 2\Delta/A$ and*

$$d_{f,g} \leq \frac{4\sqrt{M}\Delta}{\psi_{\min}A} = O(1/A). \quad (14)$$

Proof. $\alpha_{\text{sym}}^* \geq A$ gives $|a_y - b_y| \leq \max(a_y, b_y)/A \leq \Delta/A$; since $a_y - b_y = R_{\hat{y}} - R_y$, the map $y \mapsto R_y(x)$ lies within Δ/A of a constant, so $|\Delta_l| = |R_{y_l} - R_{y_0}| \leq 2\Delta/A$. Apply Thm. 11. $\square$

Corollary 23 (Finite-order control along families). *If $\sup_p \max\{D_\alpha(p||p'), D_\alpha(p'||p)\} < \infty$ for some $\alpha > 1$, then $\alpha \leq \alpha_{\text{sym}}^*$ by Thm. 5(iii), so $d_{f,g} \leq 4\sqrt{M}\Delta/(\psi_{\min}\alpha)$. Bounded divergence at order α forces representational similarity, at a rate improving in α .*

Corollary 24 (The ceiling is finite). *For any $d_0 > 0$,*

$$\alpha_0(d_0) := \sup\{\alpha_{\text{sym}}^* : d_{f,g} \geq d_0\} \leq \frac{4\sqrt{M}\Delta}{\psi_{\min}d_0} < \infty, \quad (15)$$

so every $\alpha > \alpha_0(d_0)$ controls $d_{f,g}$ below d_0 . The existence of a finite controlling order is therefore a theorem, not a conjecture.

Remark 25 (The sandwich). Thm. 5 gives $\alpha_{\text{sym}}^* > 1$; Thm. 21 gives $\alpha_{\text{sym}}^* < \infty$ for every non-equivalent pair; Cor. 24 bounds the supremum explicitly. The controlling orders form an interval anchored at 1 from below and closed from above by a computable constant.

7 What Can and Cannot Be Certified

The three positive results of Sec. 5 and Sec. 6 are correct and their qualitative predictions hold empirically, but none of them certifies a pair of trained models. The obstructions are structural and worth stating together, because they arise from three different mechanisms.

(i) Order $\alpha \leq 1$ fails outright. Thm. 5 gives $\alpha_{\text{sym}}^* > 1$ unconditionally, so total variation, squared Hellinger, Jensen–Shannon and KL admit no modulus of control at all (Cor. 6). This is qualitative: no constant can repair it.

(ii) KL’s floor cannot be lifted. Thm. 15 is non-vacuous only for $\epsilon < \delta_m^2/64$. Because the $(M+1)$ -th largest entry of a probability vector satisfies $p_{(M+1)} \leq 1/(M+1)$, we have $\delta_3 \leq 1/3$ for every model whatsoever, capping the theorem at $\epsilon < 1.7 \times 10^{-3}$. On a CIFAR-10 ResNet18 at $M = 2$ the realised requirement is $\epsilon < 1.2 \times 10^{-31}$, against observed KL of order one. Quantile floors do not help: discarding the worst 10% of inputs buys nine orders and leaves thirteen (Rem. 18).

(iii) By contrast, the sup-log-ratio route does certify. Prop. 13 shows bound $\geq \sqrt{2\lambda_{\max}}d_{\text{SVD}}$, so non-vacuity forces $d_{f,g} < 1/(2\lambda_{\max})$. Whether that cap binds is an empirical question, and on CIFAR-10 it does not: $\lambda_{\max}$ is 1.44–3.46 over $M = 2, 3, 5$, giving ceilings 0.347–0.145 against median $d_{f,g}$ of 0.128, 0.065, 0.103. With the two refinements of Rem. 12 the bound certifies 58%, 64% and 13% of retrained pairs, against 0% under the naive constant and with no violations. So of the three obstructions only two are real; the third was a technique gap. It does, however, degrade with dimension: $\lambda_{\max} \approx 0.68M$, so the usable range is low-dimensional representations. Cor. 24 remains out of reach, needing $\alpha > 4\sqrt{M}\Delta/\psi_{\min} = 64$ on these models against a threshold near 1.35.

Remark 26 (Why this is the result rather than a shortfall). (i) is tail insensitivity written as an inequality: d_{KL} is a p -weighted average of the log-ratios while ρ_2 is their unweighted spread, and any conversion pays the ratio of the two measures. (iii) is a limitation of estimating the spectrum through $\|E\|_F$; escaping it requires bounding the singular values of $\Sigma_{z'w'}$ directly, which we flag as the main open technical problem. (i) and (ii) are properties of the divergences themselves and no constant repairs them; (iii) was a technique gap and closing it yields the paper’s one usable certificate.

7.1 A diagnostic instead of a certificate

What survives is usable, and cheap. Four quantities computed from a pair of trained models bound in advance how much any likelihood comparison can reveal:

1. δ_{M+1} , the conditional-probability floor over a diverse label subset. KL can say nothing unless the observed divergence falls below $\delta_{M+1}^2/64$. Report it beside any KL-based similarity claim.
2. $\lambda_{\max}(\text{Corr}(z_1))$. No Frobenius-route bound can certify $d_{f,g} \geq 1/(2\lambda_{\max})$, so this fixes the best dissimilarity the method could ever rule out. It grows roughly as $0.68M$ on CIFAR-10, so the certificate is a low-dimensional instrument.
3. $\Delta/\psi_{\min}$, which sets the Rényi order $4\sqrt{M}\Delta/\psi_{\min}$ that Cor. 24 would require.
4. α_{sym}^* , which says where on the Rényi scale the pair actually becomes distinguishable, and is ∞ exactly when the models are equivalent (Thm. 21).

All four cost a forward pass over a held-out set. The practical guidance that follows is to prefer orders $\alpha > 1$ over KL when comparing confident models, to treat matched loss as uninformative about matched representations whenever δ_{M+1} is small, and to read $\lambda_{\max}$ before trusting any perturbation-style guarantee.

8 Open Problem: How Large Is α_0 ?

Thm. 5 and Thm. 21 sandwich the controlling orders into $(1, \alpha_0]$, where $\alpha_0 = \sup \alpha_{\text{sym}}^*$ over non-equivalent pairs, and Cor. 23 turns any bound on α_0 into a control modulus. What remains open is the value of α_0 .

Is α_0 finite, and if so what is it?

All our searches place it near 1.3. Exhaustively over the 720 permutations at $k = 6$, order-breaking pairs reach $\alpha_{\text{sym}}^* = 1.09$ and reflections 1.11, with non-uniform norms lowering both. A factorial over shrink $\times$ angular regularity $\times$ norm profile $\times$ dimension, blocked on permutation class, tops out at 1.203; none of the four factors lifts it and angular irregularity actively depresses it. Randomised search over $M \in \{2, \dots, 12\}$ reaches 1.34 (Tab. 1), with medians at 1.00. A targeted search over reflection and reflection-composed permutations (921 configurations) reaches 1.14.

Two structural facts explain the ceiling. First, $\alpha_{\text{sym}}^* \rightarrow 1$ whenever some input approaches a decision boundary of one model while the other ranks the tied labels differently; bisectors have codimension one, so any full-dimensional input support crosses them (Fig. 1c). Second, at $M = 2$ the unembeddings carry a cyclic order, and an order-breaking permutation cannot be realised without a collision: the 720-permutation census shows $\alpha_{\text{sym}}^* = \infty$ exactly on the rotations, which are globally linear, while every order-breaking

Table 1: α_{sym}^* saturates in representation dimension. Margin floor 0.08, $d_{f,g} \geq 0.10$ required, 200 random geometries per M ; n is the number surviving both filters. Medians sit at 1.00: almost every non-equivalent pair has a critical order barely above 1.

M	2	3	4	6	8	12
n	144	199	192	182	177	183
best α_{sym}^*	1.34	1.31	1.20	1.26	1.18	1.22
median	1.004	1.002	1.001	1.001	1.002	1.002
$d_{f,g}$ at best	0.12	0.33	0.32	0.42	0.24	0.41

Table 2: Rate lemma versus exact computation, single input, $k = 7$, $\rho \leq 160$. The lemma is per-direction and verifies in both: $\alpha_{\text{fwd}}^* = 1.349915$ (bisected 1.349919, rel. err 2.8×10^{-6}) and $\alpha_{\text{rev}}^* = 1.276277$ (bisected 1.276430, rel. err 1.2×10^{-4}), so $\alpha_{\text{sym}}^* = 1.2763$.

α	branch	predicted	measured
0.5	exp	0.2866	0.2866
1.0 (KL)	exp	0.2866	0.2866
1.1	exp	0.2047	0.2044
1.3	exp	0.0409	0.0401
1.6	linear	0.3414	0.3414
2.0	linear	0.5325	0.5324
10	linear	0.8110	0.8110
∞	linear	0.8623	0.8623

permutation is pinned near 1.05 however dissimilar its representations. This is a topological obstruction, not a metric one — the ceiling is flat in the number of order violations and unmoved by non-uniform norms.

9 Experiments

Primary metric. For each divergence we report the empirical modulus $\hat{h}(\epsilon) = \sup\{d_{f,g} : D \leq \epsilon\}$, the direct counterpart of Def. 1. Failure of $\hat{h}$ to approach 0 as $\epsilon \rightarrow 0$ is evidence of non-control.

Statistics. All intervals are 95% cluster bootstrap over seeds, not pairs: n seeds give $\binom{n}{2}$ dependent pairs, so pair-level resampling is anticonservative.

9.1 Constructed models

Table 2 reports the rate-lemma check. Fig. 1 shows the ladder and the crossover scatter.

9.2 Synthetic angular classification

Data are 20,000 draws from a 2D Gaussian ($\sigma = 3$) labelled by angle into c opposite-slice classes. The embedding network is three LeakyReLU layers $2 \rightarrow w \rightarrow w \rightarrow w \rightarrow 2$; the unembedding is a free $c \times 2$ table. Adam, batch 128, 3000 steps, 6 seeds per cell; all models exceed 96% accuracy. Statistics are over the 15 seed pairs per cell.

Table 3: Mean m_{CCA} against width, after applying the $\geq 90\%$ accuracy filter of [1] (315/400 models retained). Falling m_{CCA} means rising dissimilarity. n is pairs retained; the $c=18$ row is reported for completeness only, since the filter removes models very unevenly there.

c	$w=16$	32	64	128
4	0.741	0.689	0.692	0.681
6	0.639	0.597	0.562	0.533
10	0.733	0.559	0.545	0.459
18	0.833	0.625	—	0.490

pairs retained, $c=4$: 190 at every width; $c=6$: 190 except 171; $c=10$: 136–190;
 $c=18$: 45, 3, 0, 10, 28.

Wider networks learn closer distributions and more dissimilar representations. At 15,000 steps the undertraining of our earlier run is gone: mean training loss is flat in width within each class count (e.g. $0.0202 \rightarrow 0.0208$ at $c=4$, $0.134 \rightarrow 0.116$ at $c=10$), so width and fit quality are no longer entangled. In that regime the two quantities move in *opposite* directions. Mean d_{KL} falls with width — $5.06 \rightarrow 1.65$ at $c=10$, $0.116 \rightarrow 0.065$ at $c=4$ — while mean $d_{f,g}$ rises monotonically: $0.251 \rightarrow 0.366$ ($c=4$), $0.475 \rightarrow 0.558$ ($c=6$), $0.436 \rightarrow 0.697$ ($c=10$), $0.425 \rightarrow 0.733$ ($c=18$). Regressing $d_{f,g}$ on $\log w$ with mean loss as a covariate leaves the slope positive in every case ($+0.034$, $+0.028$, $+0.098$, $+0.073$), so this is not a fit-quality artefact.

This is a direct instance of the paper’s thesis on trained models and at scale: 3,800 pairs in which increasing capacity makes the output distributions closer and the representations further apart.

The effect is not an artefact of our invariance class. $d_{f,g}$ is invariant to a strictly narrower group than the CCA-based measures used in prior width studies, so we repeated the sweep under m_{CCA} (Tab. 3). It falls monotonically with width in every class count — and m_{CCA} is a *similarity*, so falling m_{CCA} is rising dissimilarity, the same direction as $d_{f,g}$. Both measures therefore disagree with the expectation that larger models converge toward linear equivalence [1].

Scope, and what the filter does. Two restrictions matter. First, we vary hidden width at *fixed* representation dimension $M = 2$ and fixed dataset size, whereas [1] increases capacity and data together; our result speaks only to the width slice. Second, the accuracy filter is applied unevenly across cells: at $c=18$ it leaves three pairs at $w=32$ and none at $w=64$, so those cells compare differently-selected subpopulations rather than different widths. We therefore rest the claim on $c=4$ and $c=6$, where the filter removes essentially nothing (190 of 190 pairs at almost every width) and the monotone decline is still present, and treat $c=10$ as

Table 4: Label smoothing raises the confidence floor and with it KL’s predictive power. $c = 6$, $w = 64$, 6 seeds, 15 pairs per row.

smoothing	δ_m	Spearman($d_{KL}, d_{f,g}$)	p
0	5.2×10^{-93}	0.550	0.034
0.05	2.6×10^{-5}	0.782	0.001
0.1	2.4×10^{-3}	0.732	0.002
0.2	2.3×10^{-2}	0.821	< 0.001

corroborating and $c=18$ as uninformative.

Control curves and non-vacuity. On 3,800 synthetic pairs Thm. 11 is violated 0 times and non-vacuous on 1.3%, with median bound 6.689 against median $d_{f,g} = 0.570$. The contrast with CIFAR-10 (58% non-vacuous, median $d_{f,g} = 0.122$) is instructive: these synthetic pairs are far more dissimilar, and Prop. 13’s cap $d_{f,g} < 1/(2\lambda_{\max})$ binds where CIFAR-10’s does not. Certification is possible exactly when the models are already close, which is the honest scope of the method.

Agreement across similarity measures is also weaker here than on CIFAR-10: rank correlations with d_{KL} are $+0.541$ ($d_{f,g}$), -0.331 (m_{CCA}), -0.302 (CKA) and $+0.276$ (Procrustes), against 0.73–0.81 in magnitude on CIFAR-10. The measures diverge most on the pairs that are most dissimilar.

Label smoothing restores KL (the causal test). Cor. 17 predicts that KL becomes informative exactly when δ_m is lifted. Training $c=6$, $w=64$ models with smoothing $s \in \{0, 0.05, 0.1, 0.2\}$ moves δ_m by *ninety-one orders of magnitude*, from 5.2×10^{-93} to 2.3×10^{-2} , and the rank correlation between d_{KL} and $d_{f,g}$ rises accordingly (Tab. 4). This is an intervention, not an association: the geometry of the task is unchanged and only the confidence floor moves.

Non-vacuity. Thm. 11 is violated on 0/150 pairs, but its bound falls below 1 on only 5.3% of them (median bound 4.09 against median $d_{f,g} = 0.318$; 8.0% non-vacuous at $c=4$, 2.7% at $c=6$). The constant is correct and the $\sqrt{\lambda_{\max}}$ form helps, but roughly an order of magnitude of slack remains before the guarantee is usable at the scale of trained models.

9.3 CIFAR-10

Forty ResNet18 models, $M = 2$, ten seeds at each of four label-smoothing levels, 20,000 steps, batch 32; 180 pairs, none dropped. Mean test accuracy 0.737 unsmoothed. Argmax agreement between paired models has median 0.776 and minimum 0.607, so roughly a fifth of inputs are ranked differently — reported because α_{sym}^* is computed on the agreeing subset.

Table 5: Temperature sweep on CIFAR-10, 45 pairs. $d_{f,g}$ is invariant by construction (max change 0.00), so the change in rank correlation is attributable to δ_3 alone.

T	δ_3	mean d_{KL}	mean $d_{f,g}$	Spearman
1	7.9×10^{-11}	0.601	0.1918	0.725
2	8.5×10^{-6}	0.409	0.1918	0.766
4	2.5×10^{-3}	0.289	0.1918	0.787
8	3.5×10^{-2}	0.183	0.1918	0.792
16	9.6×10^{-2}	0.096	0.1918	0.816

C1: the floor worsens with representation dimension.

Across 45 pairs per dimension the operative floor δ_{M+1} is 1.1×10^{-15} ($M=2$), 8.8×10^{-15} ($M=3$) and below 10^{-16} ($M=5$), so Thm. 15 is empty by roughly thirty orders throughout. Larger M requires more labels for diversity and therefore takes the floor deeper into the tail. Restricting both models to the reference’s top- k support recovers 44%/65%/89% of the full KL at $k = 2, 3, 5$ for $M=2$, rising to 67%/84%/95% at $M=5$; the $k = 1$ entry is identically zero by construction and carries no information.

KL stops tracking representational dissimilarity as M grows. The rank correlation between d_{KL} and $d_{f,g}$ is 0.725 at $M=2$ ($p < 10^{-4}$), 0.722 at $M=3$ ($p < 10^{-4}$) and 0.143 at $M=5$ ($p = 0.35$, not significant). This is an independent confirmation of the δ_{M+1} mechanism on a variable the theory was not fitted to, and it qualifies the caveat below: the regimes in which KL remains empirically informative are the low-dimensional ones, not the representative ones.

C2: a temperature intervention isolates the mechanism.

Rescaling logits by $1/T$ leaves $d_{f,g}$ *exactly* invariant, since d_{SVD} standardises its inputs, while δ_3 rises monotonically. This holds representational dissimilarity literally fixed and moves only the confidence floor. Over 45 pairs (Tab. 5), δ_3 climbs nine orders while mean $d_{f,g}$ stays at 0.1918 to machine precision, and the rank correlation between d_{KL} and $d_{f,g}$ rises monotonically from 0.725 to 0.816 ($p < 10^{-4}$ throughout).

The smoothing arm is confounded, and we report it as such. Label smoothing costs accuracy on CIFAR-10 ($0.737 \rightarrow 0.633$ at $s = 0.2$) and moves $d_{f,g}$ as well as δ_3 (0.192, 0.153, 0.053, 0.143 across $s = 0, 0.05, 0.1, 0.2$). The resulting correlations are non-monotone (0.725, 0.640, 0.338, 0.911) and we draw no conclusion from them. The temperature design exists precisely because it removes this confound; we keep the smoothing arm as a negative control.

C3: certification works, and degrades with dimension.

Thm. 11 is violated on 0/45 pairs at every dimension. With

Table 6: Non-vacuity of Thm. 11 on CIFAR-10, 45 pairs per dimension, no violations anywhere. The refined constant is the difference between certifying nothing and certifying a majority.

	$M=2$	$M=3$	$M=5$
$\lambda_{\max}$	1.442	1.904	3.458
ceiling $1/2\lambda_{\max}$	0.347	0.263	0.145
median $d_{f,g}$	0.128	0.065	0.103
median bound, naive	1.558	1.794	2.904
median bound, refined	0.931	0.764	1.226
non-vacuous, naive	0%	0%	0%
non-vacuous, refined	58%	64%	13%

the refinements of Rem. 12 it certifies 57.8%, 64.4% and 13.3% of pairs at $M = 2, 3, 5$, against 0% everywhere under the naive constant (Tab. 6). The driver is $\lambda_{\max}$, which grows roughly linearly in M (1.44, 1.90, 3.46, i.e. $\approx 0.68M$) and tightens Prop. 13’s ceiling $1/(2\lambda_{\max})$ from 0.347 to 0.145. The ceiling does not bind at any dimension we ran, but the margin closes: median $d_{f,g}$ is 37% of the ceiling at $M=2$ and 71% at $M=5$. Extrapolating $\lambda_{\max} \approx 0.68M$, certification should fail outright once typical $d_{f,g}$ exceeds $\approx 0.74/M$, which for retrained networks of this kind puts the practical limit near $M \approx 7$.

Both failure modes track dimension together. KL’s rank correlation with $d_{f,g}$ falls $0.725 \rightarrow 0.722 \rightarrow 0.143$ and the certification rate falls $58\% \rightarrow 64\% \rightarrow 13\%$ over $M = 2, 3, 5$. The first is driven by δ_{M+1} reaching deeper into the tail, the second by $\lambda_{\max}$ growing with M — different mechanisms, same direction. Low-dimensional representations are the benign case for both halves of the theory.

C4: the findings survive a change of measure. At $M=2$ the rank correlations with d_{KL} are $+0.725$ ($d_{f,g}$), -0.768 (m_{CCA}), -0.806 (linear CKA) and $+0.806$ (Procrustes), all $p < 10^{-4}$; signs differ because m_{CCA} and CKA are similarities while $d_{f,g}$ and Procrustes are dissimilarities. Pooled over $M \in \{2, 3, 5\}$ the magnitudes fall to 0.36–0.60, which reflects the dimension effect above rather than disagreement between measures: the four move together within each dimension. The conclusions are not artefacts of our invariance class.

A caveat on the worst case. At $T = 1$ the rank correlation between d_{KL} and $d_{f,g}$ is already 0.725. KL is thus empirically informative on retrained CIFAR-10 models even though Thm. 15 guarantees nothing about them. As on synthetic data, the failure our constructions realise is a worst case that training does not generically produce; what the temperature sweep shows is that the confidence floor governs *how much* information KL carries, not whether it carries any.

9.4 Language modelling

10 Related Work

11 Discussion

Limitations. (a) *Two of the three positive results remain out of reach.* Thm. 15 needs $\epsilon < 1.7 \times 10^{-3}$ at best and Cor. 24 needs $\alpha \approx 64$ against a threshold near 1.35; neither certifies a trained pair, and Rem. 14 argues the first is not repairable. Thm. 11 does certify, on 58% of CIFAR-10 pairs, but the remaining 42% and the $d_{f,g} < 1/(2\lambda_{\max})$ ceiling bound what the method can reach; the residual slack is entirely in passing from log-ratio perturbations to coordinate correlations (Rem. 12). (b) *Thm. 15 is vacuous on trained classifiers, and not repairably so.* On CIFAR-10 at $M = 2$ it requires $\epsilon < 10^{-31}$; even the best conceivable floor caps it at $\epsilon < 1.7 \times 10^{-3}$ (Rem. 18). This is the content of Cor. 17 rather than a defect, but it does mean our KL result is of theoretical interest only. The graded bound (Thm. 22) carries the practical weight. (c) *The worst case is not the typical case.* KL retains substantial empirical association with $d_{f,g}$ on trained models (Spearman 0.725 on CIFAR-10, 0.541 on synthetic data) even where our bounds guarantee nothing. Our constructions realise a worst case that training does not generically produce; what the temperature sweep shows is that the confidence floor governs how much information KL carries, not whether it carries any. (d) *Scope.* Control is proved for embeddings and unembeddings, not intermediate layers; $d_{f,g}$ has a specific invariance class, narrower than CCA-based measures, and we do not claim it is the right one; the label-diversity assumption binds when classes are fewer than the representation dimension. (e) *The synthetic arm does not apply the accuracy filter of prior work.* Ref. [1] retains only models above 90% accuracy. Our $c=18$ arm largely falls below that (20 of 100 models under 0.80), and one $c=6$ model reaches only 0.825, so the high-class-count rows are not directly comparable to published width results. Our C1 block also aggregates δ_m across class counts rather than conditioning on them. Both are corrected in the final version. (f) *CIFAR-10 is not yet run* (Sec. 9.3).

What this changes in practice.

Impact Statement

This is foundational work on when representation comparisons are meaningful. The main societal relevance is that overclaiming representational agreement from likelihood agreement can mislead interpretability and distillation pipelines.

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